

Renatto Tommasi Hernández

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Current Residence Budapest, Hungary



PROFILE

Robotics Engineer with an M.Sc. in Intelligent Field Robotics, spanning perception, SLAM, sensor fusion, and machine learning for autonomous systems. Experienced across the full robotics stack — from real-time perception and state estimation (C++17, Python, ROS2) to deep-learning models and embedded edge deployment — with hands-on work on mobile robots, manipulators, and aerial platforms. Proven record translating research into deployed, tested systems, backed by two accepted publications and a strong mechatronics foundation in hardware, CAD, and systems integration. Effective communicator and mentor, fluent in English and German (C1) and native in Spanish.

STUDIES

08/2023 - 06/2025 M.Sc. Intelligent Field Robotics Systems
Universitat de Girona and University of Zagreb

- Erasmus Mundus Joint Masters Scholar
- **Relevant Coursework:** Hands-on Localization (10.0/10), Hands-on Perception (9.6/10), Probabilistic Robotics, Machine Learning.
- Specialization in Aerial Robotics and Multi-Robot Systems
- GPA: 8,75/10, Thesis Grade: 9,1/10

Thesis: "Enhancing Indoor Mapping and Localization in Specular Rich Environments using Deep Learning and Sensor Fusion"

08/2018 - 06/2023 B.Sc. Mechatronics Engineering
Tecnológico de Monterrey (ITESM) and Hochschule Mannheim (Double Degree DHIK)

- Double Degree in Germany at the Hochschule Mannheim, part of the DHIK program for 17 months, including semester of classes, internship and thesis.
- Recipient of a Scholarship for Excellence in Leadership from ITESM
- DHIK-DAAD Scholarship Holder
- GPA: 88/100, Thesis Grade : 1,0 (German Grade)

Thesis: "Application of a force-torque sensor for the control of robotic drilling processes".

PROFESSIONAL EXPERIENCE

07/2025 - Current AI and Robotics Engineer
European Knowledge Centre (Hungary)

- **Autonomous Mobile Robot - Perception & SLAM:** Designed and implemented a real-time multi-camera visual-inertial SLAM system (C++17, ROS2) fusing four fisheye cameras with IMU via continuous-time factor-graph optimization (GTSAM) and SuperPoint deep-learning features, reducing angular drift by 95%. Core of a first-author publication accepted at IntRob'26.
- **ESA Systems Engineering (ISS Payload):** Serve as Systems Engineer for the Prime Contractor, translating strict clinical and flight requirements into robust AI architectures. Adapting a space-ready hematology analyzer for the ISS, enabling on-device AI inference in microgravity.
- **Space Health AI (ESA Argonaut / HUNOR Program):** Lead AI researcher developing an on-device physiological health indicator for astronauts under extreme hardware constraints. Engineered Transformer-based models processing multimodal time-series data from Astroskin telemetry. (Abstract accepted in IAC 2026).
- **Foundation Models for Geospatial Data:** Deployed a stitching artifact detector utilizing DinoV3, reducing spatial data cleaning time by 50% (Transitioned to production by Airbus Geo).

01/2026 - Current Industrial Supervisor for M.Sc. Theses in Robotics
European Knowledge Centre (Hungary)

- **Topic: Visual Scene Understanding & Localization:** Directing research on bridging the gap between high-level semantics (VLMs) and

metric localization. Guiding the implementation of perception pipelines to enhance robot autonomy in unstructured environments.

- **Topic: Kinematic Path Planning:** Supervising the development of navigation algorithms for **kinematically constrained** mobile bases. Overseeing the formulation of control strategies for safe traversal in cluttered, dynamic environments.

02/2025 - 06/2025

Intern - Robotic Perception & Localization

European Knowledge Centre (Hungary)

- **Sensor Noise Rejection:** Developed a Support Vector Machine (SVM) model to eliminate reflection artifacts from Intel RealSense depth data, enhancing map quality in specular environments.
- Reduced angular drift by 95% by implementing a multi-camera Visual Odometry using 4 Fisheye Cameras, SuperPoint DL and graph optimization using the GTSAM Library.

04/2022 - 03/2023

Intern & Bachelorand - Robotic Automation

Robert Bosch Power Tools GmbH (Germany)

- **Robotic Manipulation:** Engineered an automated testing workbench using a UR16e robotic arm, integrating kinematic control with sensor feedback to simulate various operational scenarios.
- **Sensor System:** Designed and implemented a multi-modal sensing system (Force-Torque, Accelerometer, Current) for the real-time analysis of dynamic forces.
- Designed and built and testbench for automated testing of drillhammers using a UR16e.

PROJECTS

2026 Third place on the EP PerMed Hackathon 2026

- Developed an AI for Cognitive Decline on Cancer Patients due to medication toxicity. The app uses facial features to predict onboard and in a safe manner the patients cognitive baseline and degradation from the patients phone.
- Pitched the idea in front of a highly qualified jury and the other 24 participant teams.

2024 Aerial Robotics & Swarm Coordination

- Deployed ORB-SLAM2 for robust, purely vision-based state estimation and localization on a Tello UAV in GPS-denied environments.
- Engineered decentralized multi-robot coordination for a swarm of Crazyflie nano-quadrotors, successfully implementing consensus protocols and Reynolds flocking rules for dynamic obstacle avoidance.

2023 Multi-Object Tracking using Kalman Filter on Event-Based Data

- Implemented a high-frequency Kalman Filter for trajectory and velocity estimation utilizing asynchronous data streams from Dynamic Vision Sensors (DVS).
- Specifically engineered the data pipeline to handle the structural challenges of continuous, microsecond-resolution events compared to standard synchronous frames, achieving low-latency tracking.

2023 Programming of autonomous implementation onboard a Turtlebot 3

- Localization using Landmark based Graph SLAM approach and GTSAM Library
- Frontier Based Exploration using RRT* with dynamic constraints and pure pursuit PID control
- Intervention using task priority control

2018 - 2021 Project Leader for Science and Technology Inspiration @ Tec de Monterrey Guadalajara

- **Research Dissemination:** Founded and organized the **STEM Talks 2021 Forum**, a platform for students and professors to present research achievements and technical innovations.
- **Leadership & Mentoring:** Directed the organization of major robotics competitions (WER, FIRST FLL) and mentored student teams, fostering technical skills in the next generation of engineers.

2017 - 2023 **FIRST Robotics Competition (FRC)**

- Robot inspector: ensure that the 50 industrial category robots comply with the current year's FRC event regulations.
- Founder, former captain and mentor of the Prepa Tec Daedalus 7102 team

PUBLICATIONS

R. Tommasi Hernández, Dr. G. Vasilevic and Prof. Z. Istenes. "Robust Multi-Camera Visual Odometry in Specular Indoor Environments Using Deep Learning," in *Proc. IntRob26*, 2026. **(Accepted for publication)**

R. Tommasi Hernández, Dr. G. Bohus, Dr. J. Nagy-Bozsoky , P. Cordero, Z. Horvath, Dr. K.V. Nagy and Prof. B. Merkely], "Predicting Hemodynamic Shifts in Microgravity: A Deep Learning Approach to Cuffless Blood Pressure Estimation," in *Proc. IAC2026*, 2026. **(Accepted for publication)**

LANGUAGES

Spanish: Native • **English:** C1(TOEFL IBT) • **German:** C1 (Goethe Institut)

SKILLS

Core Competencies	Perception & Computer Vision, Visual-Inertial SLAM & State Estimation, Sensor Fusion (Camera, LiDAR, RGB-D, IMU), Multi-Object Tracking, Machine Learning for Robotics, Event-Based Vision.
Programming & Libraries:	C++ (≥17), Python, ROS 2, PyTorch, OpenCV, GTSAM (Factor Graphs), NumPy / SciPy.
Systems & Deployment	Embedded / Edge AI (NVIDIA Jetson), Real-Time Systems, Docker, CI/CD, Linux, CAD (CATIA, SolidWorks), Sensor Calibration.
Soft Skills	Research & Publication, Mentoring (M.Sc. theses), Cross-Functional Collaboration, Technical Communication.